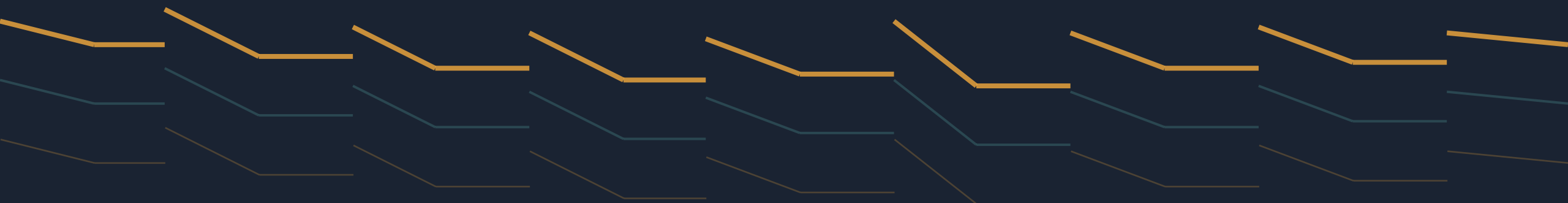


ALL - HANDS · MAY 2026

The Cognitive Cost of AI

*What's underneath the productivity gains —
and what it costs to stay in control of our own thinking.*



Robert Peake

Next Action Associates

THE SETUP

Promise vs. reality

Agentic coding makes the headlines. The rest of knowledge work lives in a different reality.

THE PROMISE

"AI is transforming knowledge work"

- Agents writing whole features
- Reports drafted in seconds
- Research compressed to minutes
- 10× productivity headlines

THE REALITY

Most people are doing AI theatre.

- Hobbled Copilot in the corner of Word
- Token cost climbing, output uneven
- Hidden cognitive debt accruing
- Quiet burnout from constant checking

Two flavours of disappointment

The headlines are about coding agents. Most people aren't coders — and most coders aren't using agents.



N O N - T E C H K N O W L E D G E W O R K E R S

AI effectiveness theatre

A hobbled Copilot bolted onto Office. Demo-grade outputs, vague gestures at "using AI," no real workflow change. The promise didn't arrive; the performance did.

"We're using AI" — for what, exactly?



E N G I N E E R S & T E C H N I C A L U S E R S

The hidden price tag

Agents work — fast — but require active piloting. Context, review, and judgement scale with output. Tokens cost money. Shortcuts accrue debt that lands later.

Productive — at a price most don't see yet.

THE CORE PROBLEM

The ragged frontier

AI is wildly uneven — brilliant in one task, useless in the next. People calibrate on the wrong slice.

WORKS BRILLIANTLY ↑



↓ FAILS QUIETLY

Best moment, worst moment

One task delights; the next one fails in a way you almost miss.

Wrong calibration

Try AI on a bad-fit task once, conclude it's useless. Or trust a hit, get burned later.

Skill is in the seam

Knowing where the frontier sits — for this task, today — is the actual literacy.

Three forces working on us

What we keep, what we carry, what we owe.



WHAT WE KEEP

Cognitive sovereignty

Staying the one who decides — what's true, what's good enough, what to ship.



WHAT WE CARRY

Cognitive load

AI moves fast; we still have to check. The burden of deep thought has shifted, not disappeared.



WHAT WE OWE

Cognitive debt

Skipped context now compounds into mistakes and dependence later. Code-debt logic, applied to thinking.

Cognitive sovereignty

Staying the author of your own thinking — even when something else is doing a lot of the typing.



THE PRACTICE

Meta-awareness.

Knowing what you're doing, why, and whether the AI's answer holds up. Without it, you become a courier for outputs you can't defend.

WHAT IT EARNS YOU



You spot the hallucination

Because you know what 'correct' looks like for this task.



You see diminishing returns

And know when to stop iterating with the model and start thinking.



You learn how to work with AI

Instead of being driven by it. Tooling becomes a craft, not a crutch.



You stay in control

The decision, and the responsibility for it, stay with you.

Cognitive load

The work didn't disappear. It moved.



THE NEW DEFAULT

Everyone is a manager now.

You're directing, reviewing, correcting, and deciding — at machine speed. The doing is faster; the thinking still moves at human pace.

WHAT'S SHIFTED

Quick wins →

No more easy answers — the easy work is automated; only the hard parts are left.

Solo deep work →

Continuous review of outputs you didn't fully draft.

Predictable pace →

Machine speed in, human judgement out — every minute.

Done means typed →

Done means verified. Checking is the new bottleneck.

Burnout is real — and quiet. It looks like productivity right up until it doesn't.

Cognitive debt

Borrowed from your future self — at compound interest.



THE ANALOGY

Tech debt: shortcuts in code that compound until the system breaks. **Cognitive debt:** shortcuts in thinking that compound until the work — or the worker — breaks.

Skipped context

You let the model fill gaps you should have. The mistake hides in those gaps.

Drift into dependence

Skills you stop exercising atrophy. You forget how to do the thing — or whether it was right.

Compounding mistakes

A small wrong assumption gets reinforced across iterations until it ships.

If we don't supply the context — because we don't have it — who will?

FROM THE FIELD

What it actually feels like

Running multi-agent AI at full tilt — the honest report.

THE OUTPUT

3×

Real work shipped — drafts, code, research, decisions. The leverage is genuinely there.

THE LOAD

~3×

If you actually pay the cost — keep sovereignty, avoid debt — the cognitive bill scales with the output.

WHAT IT FEELS LIKE

Addictive — fast feedback loops keep pulling you in.

Exhausting — constant context switching, no idle minutes.

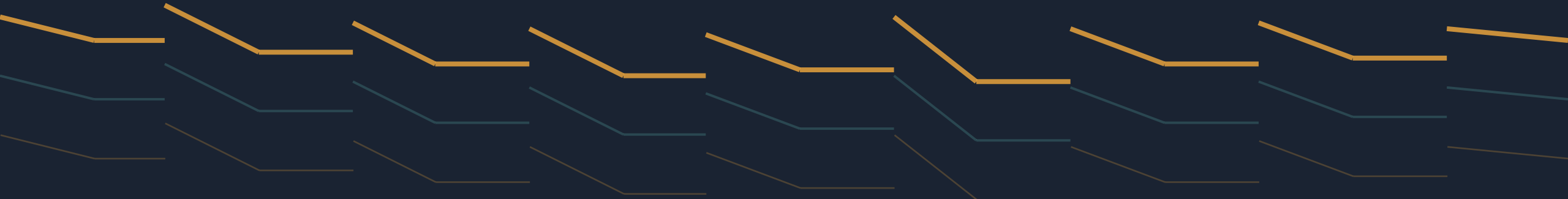
Demanding — agents asking for permissions, clarifications, decisions.

Compressed — deep work that used to take a day fits in an hour.

THE TAKEAWAY

Productivity isn't the story. *Effectiveness is.*

(And if your definition of “*Effectiveness*” doesn't include the **human**, you're **cooked**)



Robert Peake